

Synopsis: Nonsingular Gravitational Collapse via Hysterical Instability

Hysterical Response Theory proves a stable finite-radius remnant

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Computational note. Proofs, exploratory experiments, and paper writing were assisted by generative AI. Mathematical theorems stated in this paper are formally verified in Lean 4 in the companion specifications, as faithful ports of the TeX arguments at the abstractions recorded there, except results explicitly tagged as *literature hypotheses* (standard theorems cited from the literature and not re-proved here) or *physical inputs* (modeling assumptions outside mathematics).

Abstract

Short guide to the nonsingular-collapse paper. HRT identifies the wake with ordinary gravity's residual response; the remnant theorem then yields nonsingular spherical collapse.

1 Bridge from HRT

Hysterical Response is the residual / off-classical part of an already-existing interaction, $F_H = -\text{Tr}(F\mathcal{L}^{-1}\mathcal{S})$ —for gravity, a leftover of ordinary gravity, not a new force. Neutrality does not kill gravitational F_H ; closed-loop gain crossing unity makes the channel unstable.

Classical spherical gravity cancels exterior shells. That cancellation is organizational for the residual response. After horizon crossing, the interior cannot reorganize the exterior residual left in the vacated region: the wake still attracts, and relative to the collapsing surface that attraction is *outward*.

2 Wake and criticality

Theorem 2.1 (Outward Hysterical wake). *Under gravitational HRT and horizon residual-cancellation failure, the wake acceleration $H(R) = \hat{r} \cdot \mathbf{g}_H^{\text{wake}}$ is a strictly positive outward acceleration on the collapsing surface.*

Increasing curvature drives HRT criticality, so H strengthens inward and dominates GM/R^2 near the classical singular limit while remaining subdominant before the transition.

3 Remnant

Net force $F(R) = H(R) - GM/R^2$, dynamics $\ddot{R} = F - \gamma\dot{R}$.

Theorem 3.1 (Stable Hysterical Remnant). *A finite $R_* > 0$ with $H(R_*) = GM/R_*^2$ exists. If $H'(R_*) < -2GM/R_*^3$ and $\gamma > 0$, it is linearly asymptotically stable.*

4 Main claim

Theorem 4.1 (Nonsingular Gravitational Collapse). *Under spherical HRT, horizon residual-cancellation failure, curvature-driven criticality, the radial reduction, restoring slope, and positive damping, spherical collapse ends at a stable remnant $R_* > 0$ rather than $R = 0$. In that sense, Hysterical Response Theory proves nonsingular gravitational collapse.*

The load-bearing geometric input is horizon residual-cancellation failure; the wake identification and instability engine are HRT.